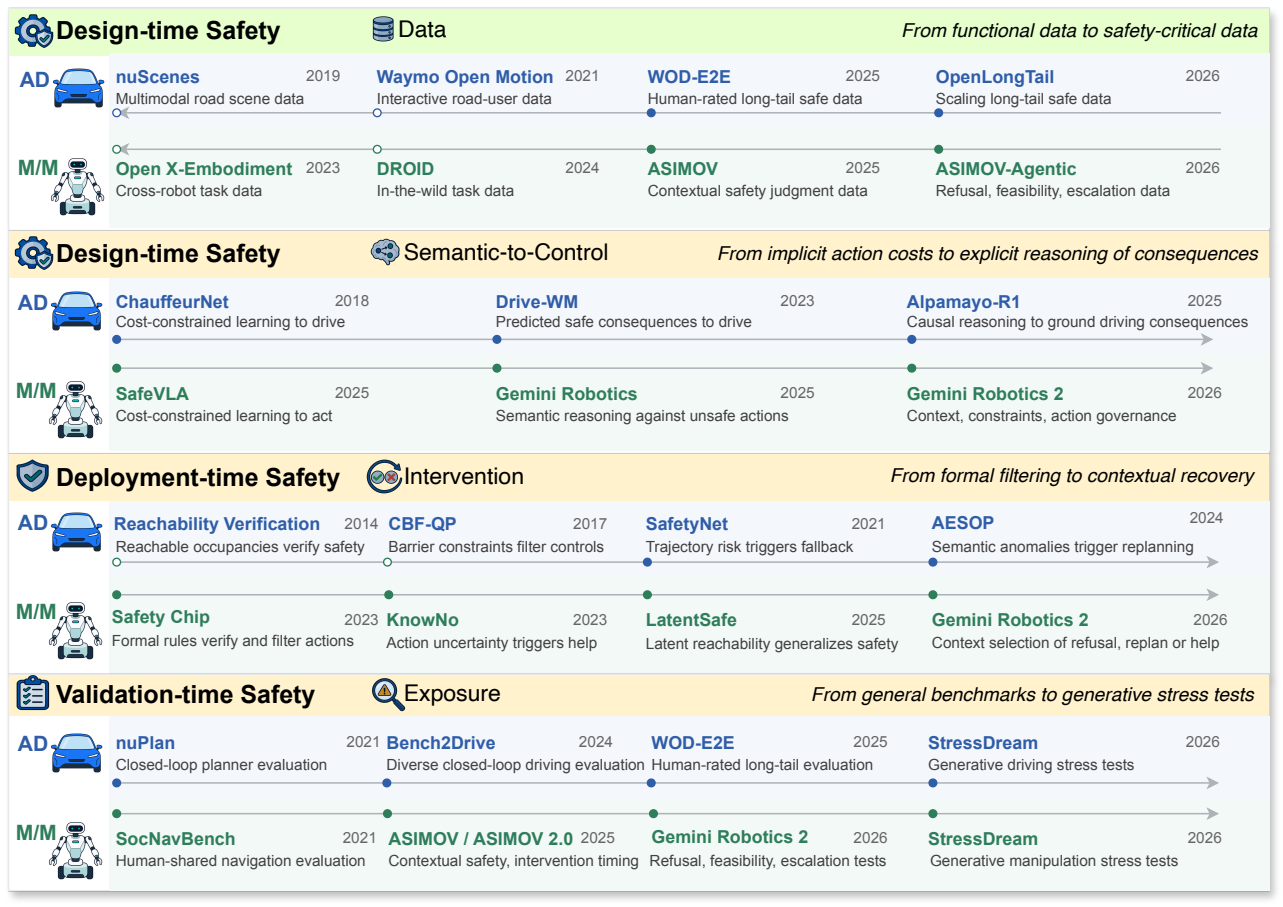

Design-time Safety
Data
From functional data to safety-critical data
AD
nuScenes 2019
Multimodal road scene data
Waymo Open Motion 2021
Interactive road-user data
WOD-E2E 2025
Human-rated long-tail safe data
OpenLongTail 2026
Scaling long-tail safe data
M/M
Open X-Embodiment 2023
Cross-robot task data
DROID 2024
In-the-wild task data
ASIMOV 2025
Contextual safety judgment data
ASIMOV-Agentic 2026
Refusal, feasibility, escalation data
Design-time Safety
Semantic-to-Control
From implicit action costs to explicit reasoning of consequences
AD
ChauffeurNet 2018
Cost-constrained learning to drive
Drive-WM 2023
Predicted safe consequences to drive
Alpamayo-R1 2025
Causal reasoning to ground driving consequences
M/M
SafeVLA 2025
Cost-constrained learning to act
Gemini Robotics 2025
Semantic reasoning against unsafe actions
Gemini Robotics 2 2026
Context, constraints, action governance
Deployment-time Safety
Intervention
From formal filtering to contextual recovery
AD
Reachability Verification 2014
Reachable occupancies verify safety
CBF-QP 2017
Barrier constraints filter controls
SafetyNet 2021
Trajectory risk triggers fallback
AESOP 2024
Semantic anomalies trigger replanning
M/M
Safety Chip 2023
Formal rules verify and filter actions
KnowNo 2023
Action uncertainty triggers help
LatentSafe 2025
Latent reachability generalizes safety
Gemini Robotics 2 2026
Context selection of refusal, replan or help
Validation-time Safety
Exposure
From general benchmarks to generative stress tests
AD
nuPlan 2021
Closed-loop planner evaluation
Bench2Drive 2024
Diverse closed-loop driving evaluation
WOD-E2E 2025
Human-rated long-tail evaluation
StressDream 2026
Generative driving stress tests
M/M
SocNavBench 2021
Human-shared navigation evaluation
ASIMOV / ASIMOV 2.0 2025
Contextual safety, intervention timing
Gemini Robotics 2 2026
Refusal, feasibility, escalation tests
StressDream 2026
Generative manipulation stress tests